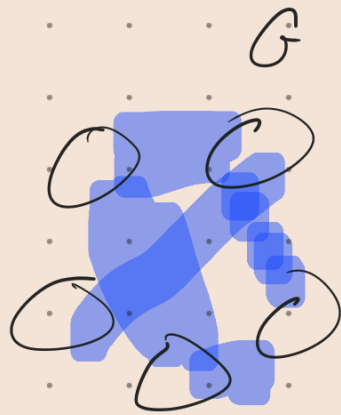


Szemerédi's Regularity Lemma

Roughly speaking: SzRL says that

"every graph can be partitioned into bounded # parts s.t. between most of pairs of parts, the graph induced is pseudorandom"



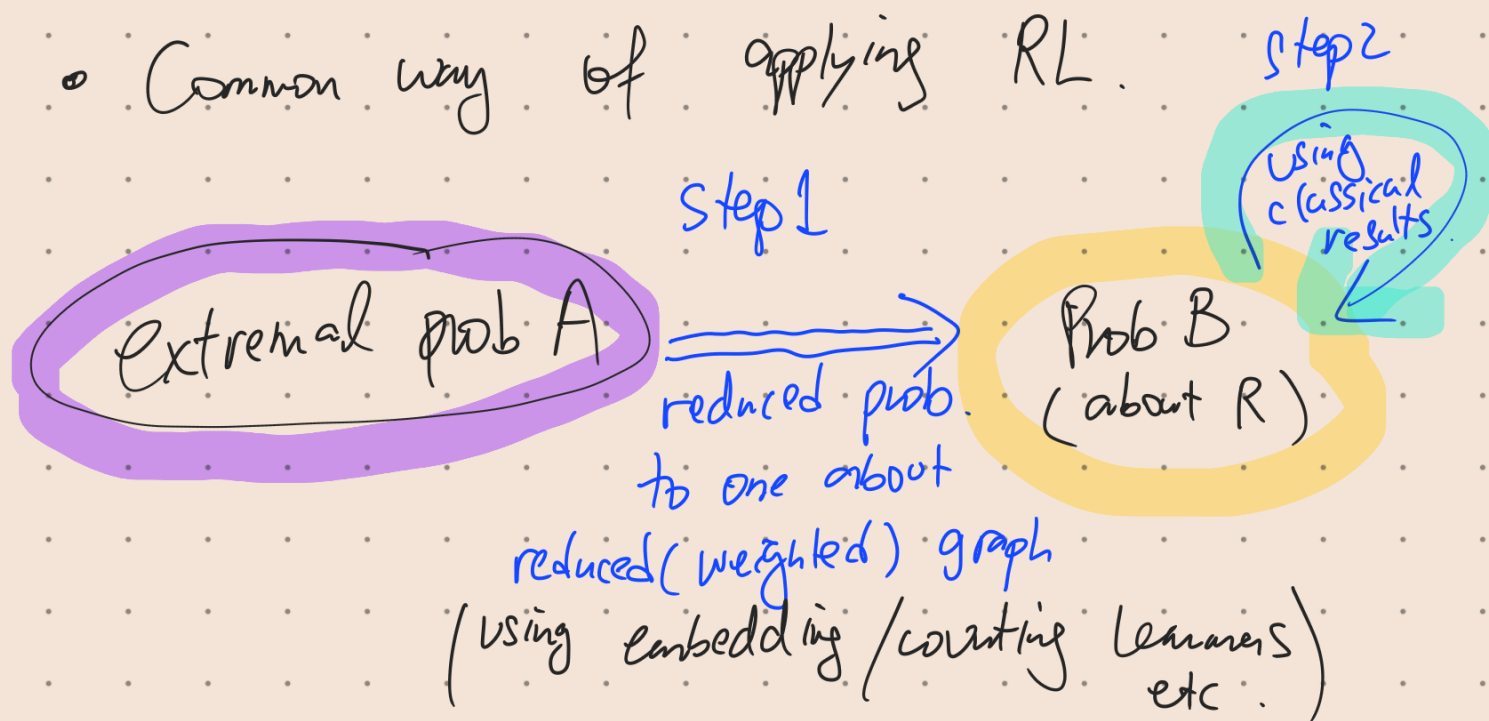
Essence of RL is

Approximating large structures

by (small) structure with low complexity

(one example: approximate subgraph densities)

• Common way of applying RL.



Thm ESS $\forall H, ex(n, H) = \left(1 - \frac{1}{\chi(H) - 1} + o(1)\right) \binom{n}{2}$

Lem (Embedding Lem) Let H be a graph. For large n

$$\bullet G \begin{cases} n = |V| \\ H\text{-free} \\ \text{edge density } \tau \end{cases} \rightsquigarrow \exists R \begin{cases} H\text{-free} \\ e(R) \geq (\tau - o(1)) \binom{|R|}{2} \end{cases}$$

reduced graph

Pf (upp bd). $\exists$ homomorphism $H \rightarrow K_{\chi(H)}$

$\Rightarrow R$ is $K_{\chi(H)}$ -free as G is H -free

$$\Rightarrow o(1) + \tau \leq \frac{e(R)}{\binom{|R|}{2}} \leq 1 - \frac{1}{\chi(H) - 1} \quad \square$$

Turán Thm

P1: $\forall$ fixed H , the subgraph density of H

$$\left(\frac{\# \text{copies of } H}{\binom{n}{|H|}} \right)$$

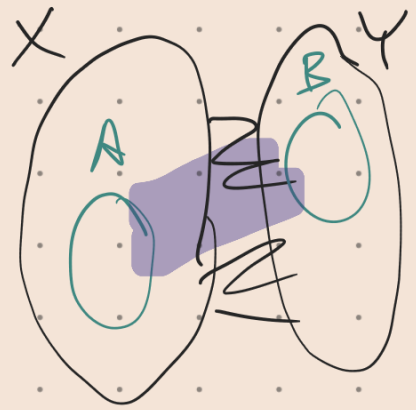
P2: R has homo. image of $H \Rightarrow G \geq H$

————— Setup —————

ϵ -regular pair: $G = (V, E)$, two disjoint sets $X, Y \subseteq V$, $e(X, Y) = e(G[X, Y])$

the density between X, Y is

$$d(X, Y) = \frac{e(X, Y)}{|X||Y|}$$



For $\varepsilon > 0$ (precision parameter)

(X, Y) is ε -regular if

$$\forall A \subseteq X, \forall B \subseteq Y \text{ with } \begin{cases} |A| \geq \varepsilon |X| \\ |B| \geq \varepsilon |Y| \end{cases}$$

then $\Rightarrow |d(A, B) - d(X, Y)| < \varepsilon$

We say (X, Y) is (ε, δ) -regular

if in addition $d(X, Y) \geq \delta$.

• ε -reg means that edges are uniformly distributed.

Def: (Regular Partition) A partition

$V = V_0 \cup V_1 \cup \dots \cup V_r$ is ε -regular

if (i) $|V_0| \leq \varepsilon |V|$ (exceptional set)

(ii) $|V_1| = |V_2| = \dots = |V_r|$

(iii) all but $\leq \varepsilon r^2$ pairs of (V_i, V_j)

are ε -regular.

Remark 1) usually V_0 is
larger than V_i , $i \in [r]$

2) edges inside each V_i
are negligible ($\sum e(G[V_i]) = o(e(G))$)

3) Other version (reg. partition)

• no exceptional set V_0
& • $|V_i| - |V_j| \leq 1 \quad \forall i, j \in [r]$

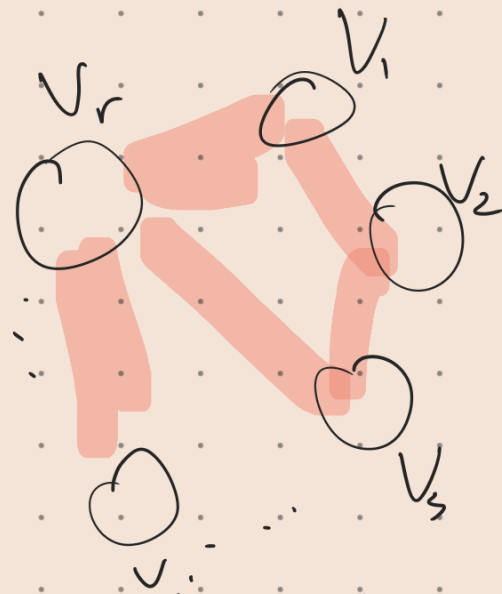
4) For (ii), we can further require
that $\forall i \in [r]$, $\leq \varepsilon r$ pairs involving
 V_i are irregular.

Thm (Szemerédi Reg. Lem 76)

Given $\varepsilon > 0$ & $m \in \mathbb{N}$, then there exists

$M = M(\varepsilon, m)$ s.t. $\forall$ graph G ,

$\exists$ an ε -reg. partition $V = V_0 \cup V_1 \cup \dots \cup V_r$
with $m \leq r \leq M$.



Rank

• ϵ usually taken to be small

$$\epsilon = o(1)$$

• Both $m \leq r \leq M$
 $\uparrow$
 meaningful

$m > 1/\epsilon$ so that

$$\Rightarrow 1/r < \epsilon$$

$\Rightarrow$ edges in V_i
negligible.

$$|V_i| \leq \frac{n}{r} \leq \epsilon n$$

edges in V_i

$$= O(\epsilon^2 n^2)$$

• M is needed (e.g. counting lem)

• Sz RL deals with graphs with positive edge density

height $\epsilon^{-1/6}$ • M from pf of Reg Lem height ϵ^{-5}
 $\left\{ 2^{2^{\dots 2}} \leq \dots \leq 2^{2^{\dots 2}} \right\} \rightarrow 2\epsilon^{-5}$

• The ϵr^2 irregular pairs are needed.

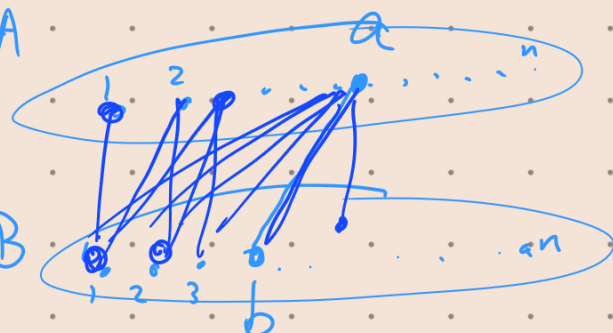
Ex Half graph

$$a \sim b$$

$$\Leftrightarrow a \geq b$$

$$[n] = A$$

$$[n] = B$$



There are ϵ -proportion of irreg pairs in any part.

Between a regular pair, the following lemma says that, for most of x s, they have the "correct" degree towards the other side.

Lem Let (X, Y) be an ε -regular pair with density d . Let $B \subseteq Y$ with $|B| \geq \varepsilon |Y|$, then all but $2\varepsilon |X|$ x s in X have degree $(d \pm \varepsilon) |B|$ in B .

$$d - \varepsilon \leq \cdot \leq d + \varepsilon$$

Pf: Let A be the set of atypical x s.

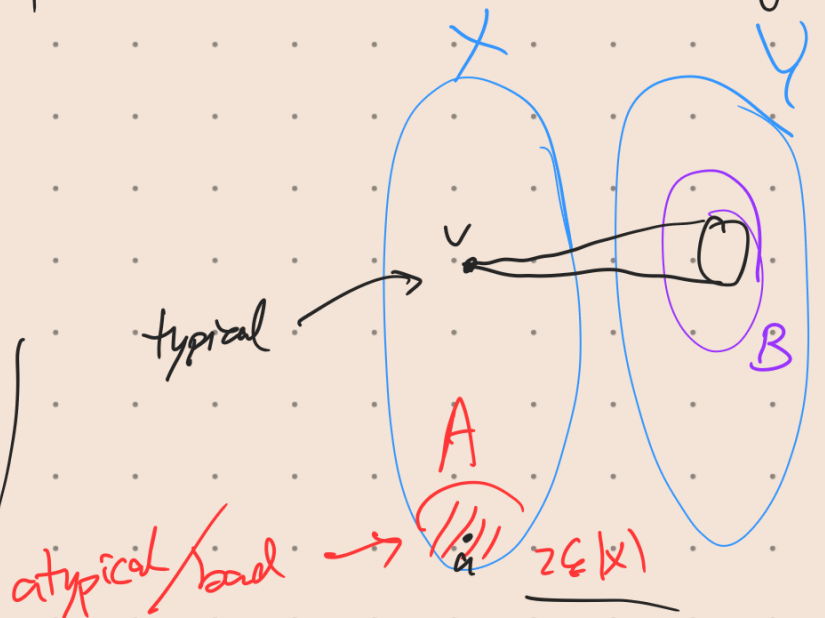
i.e. $\deg(a, B) > d + \varepsilon$

or $\deg(a, B) < d - \varepsilon$

Say w.l.o.g. that $A = \{x \text{ s of too large } \deg\}$

$$|A| \geq |A'|/2 \geq \varepsilon |X|$$

$$\Rightarrow d(A, B) > d + \varepsilon \quad \left(\begin{array}{l} \varepsilon\text{-reg} \\ |d(A, B) - d(X, B)| \leq \varepsilon \end{array} \right)$$



Given a reg. pair (X, Y) , we also know that almost all pairs of x 's from one side have the "correct" co-degree to the other side.

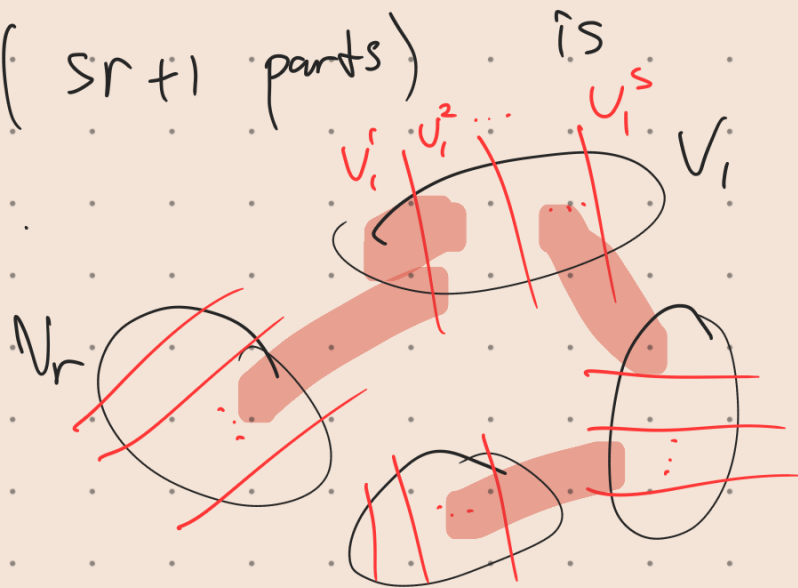
Exer: Formulate the above statement rigorously & prove it.

- The next lemma says that regularity is inherited by large subsets of pairs.

Lem (Slicing Lemma) Let $V_0 \cup V_1 \cup V_2 \cup \dots \cup V_r$ be an ε -reg. partition. Further refine each part into s equal parts: $V_i = V_i^1 \cup \dots \cup V_i^s$

Then the new partition ($sr+1$ parts) is

$\bigcirc (s \cdot \varepsilon)$ -regular.



Exer: prove the Slicing Lemma.